

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5
Marks:25

Total

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Priyadarshini K.S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Priyadarshini K.S
Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	35 Marks	Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls)	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answers	25 Marks	All answers are correct, precise, and relevant	Mostly correct with minor errors	Partially correct with noticeable errors	Mostly incorrect or irrelevant
Application of Knowledge	15 Marks	Effectively applies concepts to scenarios/questions	Applies with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15 Marks	All parts of each question are fully answered	Most parts covered with small omissions	Some parts missing	Incomplete answers
Clarity & Presentation	10 Marks	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Marks Evaluation:

Question No.	Conceptual Understanding (30%)	Accuracy of Answers (25%)	Application of Knowledge (20%)	Completeness of Response (15%)	Clarity & Presentation (10%)	Total Marks
Q1						
Q2						
Q3						
Q4						
Q5						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Name: Priyadarsini

Reg: 192511248

Sub: C8A 0404

Code: OS.

CO3 - ATI - Short Answer Analysis

1) A modern OS supports multiple users & processors with shared resources like CPU, Memory, & I/O devices.

- Classify diff. types of OS & justify.
- Draw structure of OS showing resources.
- Write sequence of system calls req. to create new process.

2) Consider OS managing multiple files & devices for concurrent users.

- Categorise system calls
- How OS handles file operation
- Pseudo code call sequence [open, read]

3) Assume real-time OS used in embedded system.

- Type of OS & explain why?
- Diagram interaction b/w hardware
- List & explain system calls.

4) Distri. computing system consists of multiple computers.

- a) Type of OS & compare. with control.
- b) Architecture showing node communication.
- c) Sequence of system calls.

5) Scenario where OS must manage process scheduling

- a) Major function of OS
- b) Block diagram of components.
- c) System call for memory allocation.

Solutions:

1) a) Types of OS:

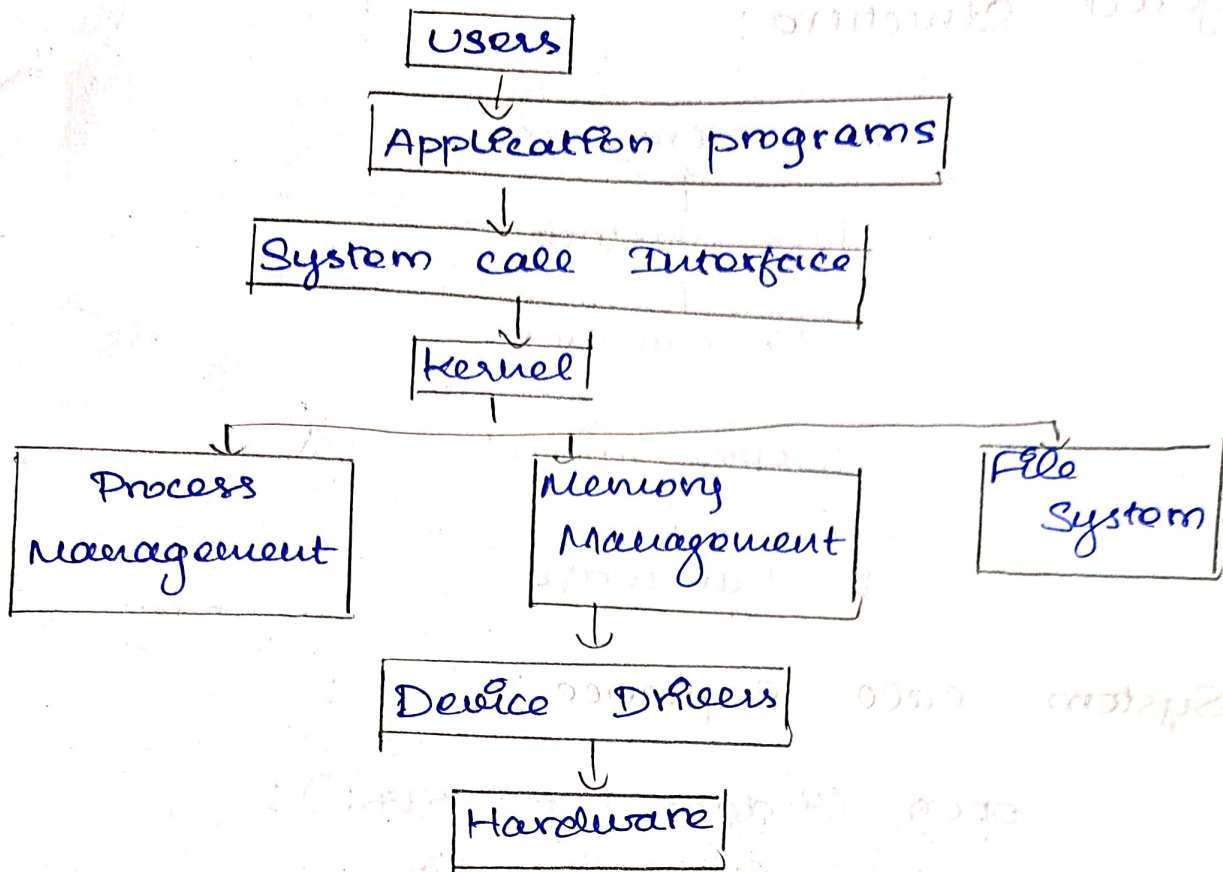
Multi-user OS - Allow multiple users to access system simultaneously.

Multi-tasking OS - Runs multiple processes simultaneously.

Time-sharing OS - Shares CPU time among users efficiently.

Justify: These OS types provide efficient resource sharing, fast response.

b) OS Structure :



c) System call Sequence :

`fork()` → creates new process
`exec()` → executes new program
`wait()` → parent waits for child
`exit()` → Terminate process.

2) a) File & Device Management System calls :

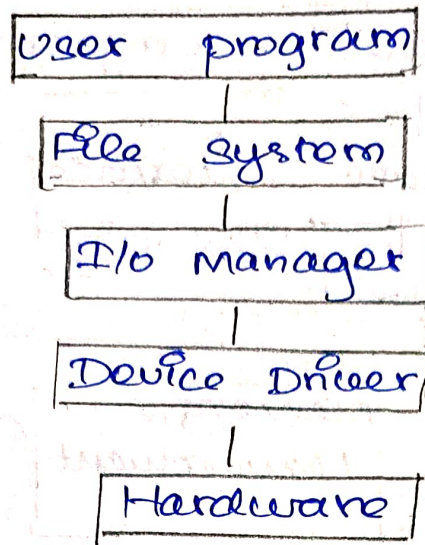
File management

`create()`
`open()`
`read()`
`write()`
`close()`
`delete()`

Device Management

request device
read device
write device
release device
`ioctl()`

b) Layered Structure :



c) System call sequence :

```
open ("data.txt", READ);  
read (file, buffer);  
close (file);
```

open() → opens the file

read() → Reads data from file

close() → closes the file.

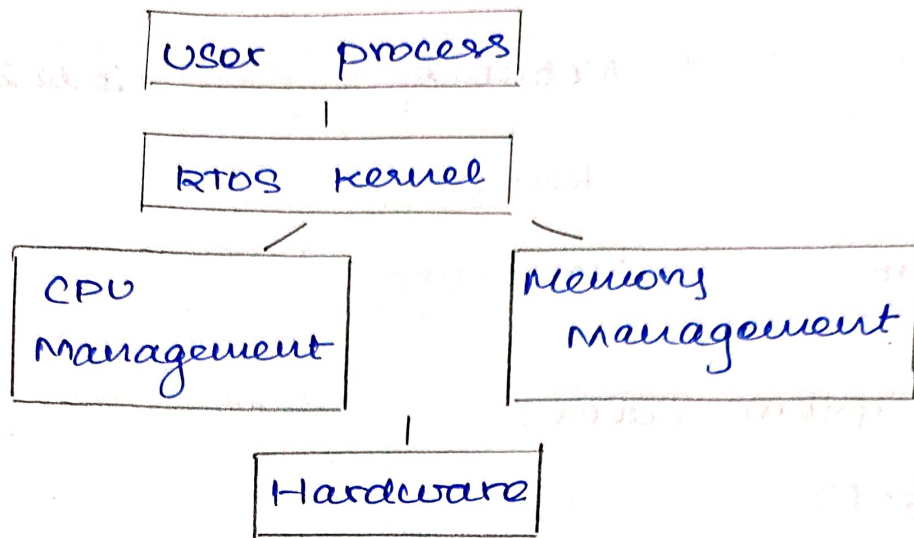
3) a) Type of OS :

Real-time Operating System [RTOS]

It is suitable because

- * Fast response
- * Predictable execution
- * Meets strict deadlines
- * High reliability.

b) Diagram



c) Synchronization System calls.

fork() - create process

wait() - wait for child process

Sem_wait() - Lock Semaphore

Sem_post() - Release Semaphore

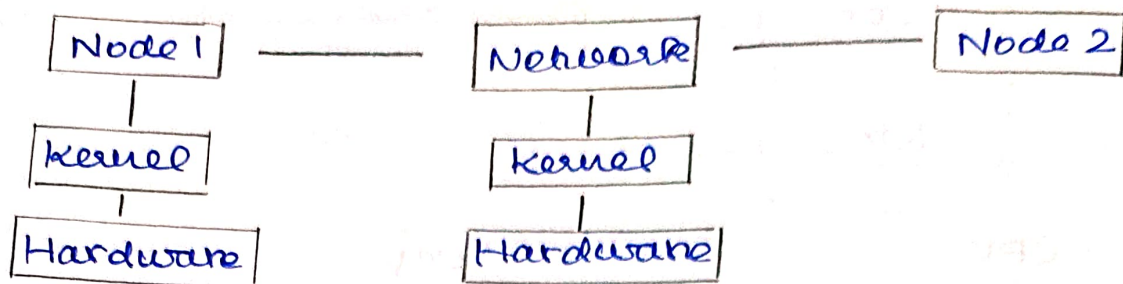
pthread_mutex_lock()

pthread_mutex_unlock()

4) a) Distributed OS :

Distributed OS	Centralised OS
Multiple computer	Single computer
Resource sharing	Local resources
Fault tolerant	Less fault tolerant
Scalable	Limited scalability

b) Architecture :



c) IPC System calls :

pipe()
fork()
write()
read()
close()
wait()

5) a) Major Functions of OS :

Process mgmt : Creates, schedules & terminate process

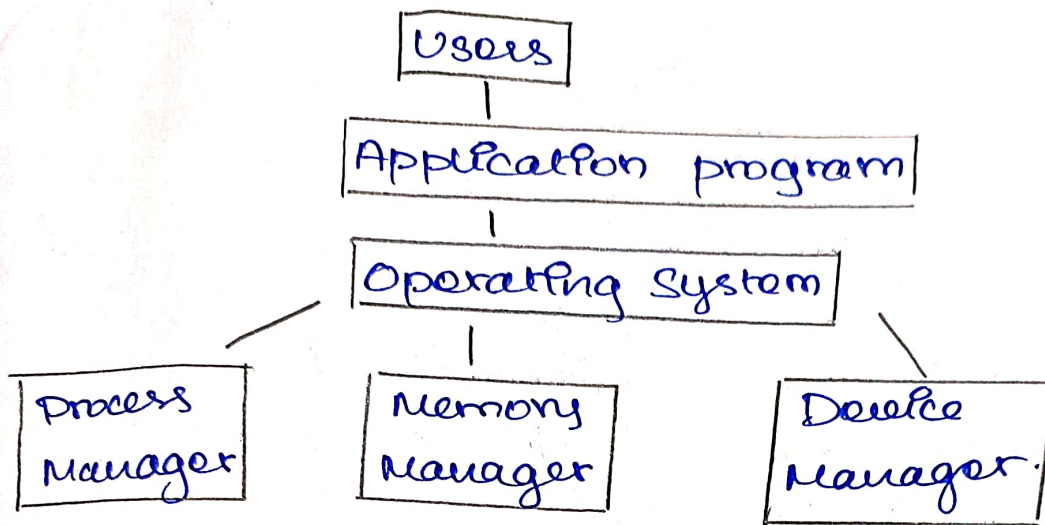
Memory mgmt : Allocates & deallocates memory

Device mgmt : Controls I/O devices through drivers.

File mgmt : Organizes files & directories

Security : protects system resources.

b) Block Diagram :



c) System call :

open()
read()
write()
close()
fcntl()